

Supplemental Material for “Roton-like Reconstruction of the Emergent Photon in Quantum Spin Ice”

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Equation numbers of the form (1), (2), ... refer to the main text; equations of this document are numbered (S1), (S2), ...

THE COMPUTED OBSERVABLE AND THE NEUTRON CROSS SECTION

Throughout, the quantity we compute is the zero-temperature, sublattice-summed longitudinal correlation function of the local Ising pseudospin,

$$S^{zz}(\mathbf{q}, \omega) = \frac{1}{N} \sum_{\mu} \sum_n |\langle n | S_{\mu}^z(\mathbf{q}) | 0 \rangle|^2 \delta(\omega - \omega_n), \quad (\text{S1})$$

with $S_{\mu}^z(\mathbf{q}) = \sum_{i \in \mu} e^{-i\mathbf{q} \cdot \mathbf{r}_i} S_i^z$ and μ running over the four pyrochlore sublattices. Here S_i^z is the component along the local $\langle 111 \rangle$ easy axis at site i , which is the component that equals the emergent electric field, $E_i = S_i^z$. All intensities quoted below — “share of the intensity,” “off-pole weight” — are fractions of $\sum_n \sum_{\mu} |\langle n | S_{\mu}^z(\mathbf{q}) | 0 \rangle|^2$ at the same $\mathbf{q}$, so they are normalized within Eq. (S1) itself.

Equation (S1) is not the neutron cross-section. The magnetic neutron cross-section of a non-Kramers or dipolar–octupolar spin ice involves, in addition, the ionic form factor, the g -tensor projecting the local pseudospin onto the laboratory magnetic dipole, the rotation from the four local $\langle 111 \rangle$ frames to the global frame, and the neutron polarization factor $\delta_{\alpha\beta} - \hat{q}_{\alpha}\hat{q}_{\beta}$. Those factors multiply Eq. (S1) with sublattice-dependent and $\mathbf{q}$ -dependent coefficients and are material specific; for the dipolar case they are given explicitly in Refs. [1, 2]. We do not apply them, because every statement we make is a comparison between eigenstates *at the same $\mathbf{q}$ and in the same sublattice channel*, where these prefactors are common and cancel from the ratios. They do not cancel from a comparison between different $\mathbf{q}$, and we therefore make no claim about the absolute momentum dependence of neutron intensity. When we write that a level “carries half the intensity” we mean half of the longitudinal pseudospin weight of Eq. (S1) at that momentum. Converting to a measured cross-section for a specific material requires the material’s g -tensor and is outside the scope of the pure ring model.

Our calculation is exact within the specified finite Hilbert spaces, not within the full microscopic spin model. Equation (1) of the main text conserves winding, and we work in the ground-state winding sector of

the spin-ice manifold. Electric charges (spinons) are absent by construction. The magnetic sector is retained, so spectral weight outside the photon branches could in principle have magnetic rather than multiphoton character; we distinguish these possibilities below. A word on which plaquettes enter Eq. (1) of the main text. A periodic cluster contains six-site closed loops of two kinds: the elementary hexagons of the infinite pyrochlore lattice, and loops that close only by winding around the torus. The latter are not plaquettes of the lattice, and including them adds ring terms with no counterpart in the thermodynamic limit; on small clusters they are the majority of all six-cycles found by a naive search. We therefore keep only loops of zero winding number, of which the 48-site cluster has exactly 48, twelve in each of the four $\langle 111 \rangle$ orientations. This is a property of the cluster geometry and is checked directly on each cluster used here; no result below depends on it being taken on faith.

CLUSTER CONSTRUCTION, SECTORS, AND NUMERICAL METHODS

Ice configurations are enumerated exactly by depth-first assignment, one site at a time, backtracking as soon as any tetrahedron would acquire a third spin of either sign, and are stored as bit masks. The ring Hamiltonian is built sparsely: for each hexagon a bit mask and the two flippable bit patterns identify the acted-on states in one vectorized pass, and the images are located by binary search in the sorted state list. The winding sector is identified from the polarization $\sum_i (n_i - \frac{1}{2}) \mathbf{r}_i$, which is conserved by Eq. (1) of the main text; we retain the sector carrying the ground state.

On the 48-site cluster the ground-state winding sector has dimension 9104 and is diagonalized completely with a dense symmetric eigensolver, so all 9104 eigenpairs are available and the operator-resolved weights saturate the equal-time sum rule $\sum_n |\langle n | S_{\mu}^z(\mathbf{q}) | 0 \rangle|^2 = S_{\mu}(\mathbf{q})$ exactly rather than to a tolerance. On the 64-site cluster the sector has dimension 249 180 and is treated by Lanczos with full reorthogonalization, the dynamical response being obtained by a continued-fraction expansion seeded with $S_{\mu}^z(\mathbf{q}) | 0 \rangle$; all 48-site benchmarks are reproduced before any 64-site number is quoted.

Going beyond 64 spins requires changing the state representation. Storing a configuration in a single 64-bit word caps the calculation at 64 sites, which is where

the published constrained-space spectroscopy has stood. We therefore store each configuration as a *pair* of 64-bit words and redefine masking, flipping, sorting and lookup on that pair. Lookup is the only part that is not immediate: the table is sorted by the pair lexicographically, and because the high word takes fewer than 2^8 distinct values at these sizes, queries are grouped by high word and each group resolved with a single vectorized binary search on the low word within its block. The result is exact, not hashed. The implementation is validated by running it on the 48-site cluster, where it reproduces the 64-bit pipeline exactly: $E_0 = -10.55601176K$, $S(L) = 0.37036$ and branch energies $1.0235K$ and $1.7303K$.

On the 72-site cluster the ice manifold contains 12 846 186 configurations in 310 winding sectors. The ground state does *not* lie in a zero-polarization sector: on this box the zero-polarization sector is empty, and the two ground sectors are a degenerate $\pm P$ pair of nonzero integer polarization, each of dimension 973 008. (An earlier version of our tooling labeled the minimum- $|P|$ sector “zero-polarization”; we flag this because the distinction matters for the Monte Carlo initialization of Appendix , which must reach this nonzero- P sector.) The identification is verified rather than assumed, by diagonalizing the largest sectors independently and comparing, which also exposes the expected exact degeneracy between the $\pm P$ pair, both at $E_0 = -14.80971536K$. The dynamical response is again obtained by continued fraction seeded with the four sublattice components of the probe. Because those four seeds produce separate lines at the same energy, they are summed by energy before any branch energy or weight is quoted; only the channel sum is basis-independent.

The 96-site cluster required two further steps. The two-billion-state ice manifold is enumerated by a vectorized frontier that imposes one tetrahedron at a time, visiting at each step the tetrahedron that introduces the fewest new sites and branching the frontier into independent seeds so that peak memory is bounded by one branch (a hundredth of a gigabyte) rather than the whole frontier; and exact state lookup is a single lexicographic **searchsorted** on the (hi,lo) word pairs, since at this size a block-by-block search over the $\sim$ millions of distinct high words is no longer viable. The winding sector is identified from an exact integer polarization key without sorting the manifold, candidates estimated from a four-million-state sample and each extracted in one masked pass; the ground sector (dimension 146 758 352, 2.7×10^9 nonzeros) is identified by diagonalizing the largest candidates and comparing. What excludes a smaller sector from lying lower: at 48 and 72 sites, where every winding sector can be enumerated and the largest ones diagonalized, the ground state always lies in a sector of maximal dimension, with sub-leading sectors higher by $O(1)K$; at 96 sites we verify this for the three largest candidates directly and rely on the same pattern for the

rest. On the cubic Monte Carlo boxes the sector reached by the seed randomization is certified by the energy gate of Appendix , which resolves a wrong sector at the percent level in E_0/N against a statistical error two orders smaller. The continued fraction uses real Lanczos on the real and imaginary parts of each probe separately, which is exact for a real symmetric H and halves the Krylov storage; 120 iterations reproduce the 250-iteration complex results on the 72-site cluster to every digit.

Momentum labels are certified by explicit translation projectors

Sector certification

Krylov convergence at 96 sites

THE GAUSSIAN BENCHMARK

Every statement of the form “about half the Gaussian-photon energy” refers to the following construction, which is quantized on the same finite torus as the exact diagonalization so that no continuum extrapolation enters the comparison.

Write $B_p = \sum_\ell C_{p\ell} A_\ell$, where C is the alternating lattice curl: $C_{p\ell} = \pm 1$ with alternating sign around the six links of hexagon p , and zero otherwise. Expanding the compact magnetic energy of Eq. (1) of the main text to quadratic order and adding the electric term gives

$$H_G = \frac{U}{2} \sum_\ell E_\ell^2 + K \sum_p B_p^2, \quad [A_\ell, E_{\ell'}] = i\delta_{\ell\ell'}. \quad (S2)$$

Restricting to momentum $\mathbf{q}$ and using the four-dimensional sublattice Bloch basis, the normal-mode frequencies are set by the singular values $\sigma_\lambda(\mathbf{q})$ of the curl in that block,

$$\omega_\lambda(\mathbf{q}) = \sqrt{2UK} \sigma_\lambda(\mathbf{q}). \quad (S3)$$

The construction is checked, not assumed. (i) Gauge invariance: $C d_0 = 0$ to machine precision, where d_0 is the tetrahedron incidence matrix signed by the up/down bipartition; the largest element of $C d_0$ is 0 exactly. If this fails the sign convention is wrong and nothing downstream is meaningful. (ii) At every $\mathbf{q} \neq \Gamma$ the block has exactly two nonzero singular values, the two transverse photon polarizations, and at Γ it has none. (iii) The Bloch bases are orthonormal, and the probe at $+\mathbf{q}$ pairs with modes at $-\mathbf{q}$, verified by projection completeness.

A single number is fitted: the overall energy scale $\sqrt{2UK}$. In the pure ring model U is generated dynamically rather than being a bare coupling, so its value is not available from Eq. (1) of the main text; this is the one place where we import a scale from the data. It is determined by one linear least-squares fit of the dominant exact line against the dominant Gaussian branch across

all momenta, giving $\sqrt{2UK} = 0.5868 K$. Everything else — the *shape* of the dispersion across momenta, the degeneracy or splitting of the two polarizations at each $\mathbf{q}$, and the distribution of one-photon weight over momenta and sublattice channels — is parameter-free geometry. The resulting values are those plotted as the dashed Gaussian curve of Fig. 1 of the main text; at L , $\omega_G = 2.03K$.

We stress the limitation this creates. Because one scale is fitted to the exact spectrum, the Gaussian curve cannot be used to test whether the overall photon velocity is renormalized: by construction it is not. It can only be used for statements that are invariant under a uniform rescaling of the whole branch — which momenta are anomalous *relative to the others*, and whether the two polarizations split. Those are the only statements we draw from it.

BASIS-INVARIANT POLARIZATION RESOLUTION

At each momentum the electric field lives in the four-dimensional sublattice space, in which the lattice curl has exactly two nonzero singular values. The neutron probe is therefore confined to a two-dimensional transverse subspace; the weight found in the remaining two directions is the longitudinal residue, which must vanish because $\nabla \cdot \mathbf{E} = 0$ in the ice manifold. It does, at the level of 10^{-31} relative to $S(\mathbf{q})$ at every momentum. This is an exact gate on the decomposition, not a fit.

A caution is necessary before any polarization statement is made. The two Gaussian transverse branches are *exactly* degenerate at every momentum, so the singular-value decomposition determines the two polarization directions only up to an arbitrary rotation within that subspace. A statement such as “this level is 68% polarization 1” is consequently meaningless in isolation. The invariant content is the geometry of the levels’ polarization vectors relative to one another. For a level n of multiplicity m we form the 2×2 polarization density matrix $M_n = A_n^\dagger A_n$, where A_n is the $m \times 2$ matrix of transverse amplitudes $\langle n, \alpha | S^z(\mathbf{q}) | 0 \rangle$. Both the purity $\text{tr } M_n^2 / (\text{tr } M_n)^2$ and the normalized overlap $\text{tr}(M_A M_B) / (\text{tr } M_A \text{tr } M_B)$ between two levels are invariant under rotations of the degenerate subspace.

The result is unambiguous. Every bright level, including multiplets of multiplicity up to four, has polarization purity 1.000000: the whole multiplet couples through a single polarization direction. And the two brightest levels at each momentum are mutually orthogonal in polarization, so a basis exists in which each is pure. They are therefore the two transverse photon descendants, identified without reference to any energy scale.

$\mathbf{q}$	ω_1/K	ω_2/K	splitting	overlap
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	1.895	1.980	0.044	2×10^{-28}
$(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$	1.744	2.069	0.171	1×10^{-29}
$(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$	2.533	2.687	0.059	4×10^{-2}
L	1.024	1.730	0.513	3×10^{-3}
X	2.235	3.423	0.420	1×10^{-31}

The splitting column is $|\omega_1 - \omega_2| / \frac{1}{2}(\omega_1 + \omega_2)$, the convention used throughout this paper; normalizing instead by the lower branch would give larger numbers for the same data. Orthogonality is exact to machine precision at $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ and X , where the little group forbids mixing, and holds to 0.3% at L and 4% at $(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$, where it does not. The residual mixing is small enough that the branch assignment is never in doubt.

Two consequences matter for the interpretation. First, the identification of the anomalous L level as a photon descendant, and not a new species, rests on this decomposition and is independent of the fitted Gaussian scale. Second — and this must be stated plainly — the “interaction-induced polarization splitting” and the “roton-like softening” at L are not two independent findings. They are the same fact viewed twice: the lower of the two transverse branches at L is the anomalous level, and the splitting between the branches is the amount by which it has been pushed down. We report the splitting because it connects to a specific earlier prediction [3, 4], not as separate evidence.

THE ROKHSAR–KIVELSON DEFORMATION AND THE CENSUS

classify the eigenstates we use the conventional Rokhsar–Kivelson extension of the pyrochlore ring model,

$$H(V) = -K \sum_p (W_p + W_p^\dagger) + V \sum_p n_p^{\text{flip}}, \quad (\text{S4})$$

where n_p^{flip} projects onto an alternating hexagon, so that $\sum_p n_p^{\text{flip}}$ simply counts the flippable plaquettes in a configuration. The sign convention matters for reading the figures and we state it once: $V > 0$ penalizes flippable plaquettes, $V < 0$ rewards them, and the exactly solvable Rokhsar–Kivelson point sits at $V = +K$, where the equal-amplitude superposition is an exact ground state. Our scans run over $V/K \in [-0.5, +0.5]$ unless stated otherwise, and the anomalous level softens toward negative V , that is, toward configurations with more flippable plaquettes. This flippability potential was introduced for the pyrochlore ring model in the original gauge-theory construction and subsequently used to map the quantum-ice phase diagram and tune the emergent electrodynamics [5–8]. Our use of it is different. We remain at the physical

ring point $V = 0$ and use only the infinitesimal derivative of an individual excitation energy. Hellmann–Feynman gives

$$\frac{d\omega_n}{dV} = \langle n | \sum_p n_p^{\text{flip}} | n \rangle - \langle 0 | \sum_p n_p^{\text{flip}} | 0 \rangle, \quad (\text{S5})$$

so the slope is an exact state-resolved census of the number of flippable hexagons added or removed by the excitation. No parameter is fit to obtain this diagnostic.

At L the two strongest S^z -active levels respond with opposite signs. The level at $1.73K$, close to the Gaussian-photon energy $2.03K$, loses flippable hexagons as V is increased and behaves as an ordinary dressed photon. The lower level at $1.02K$ carries 51% of the S^z spectral weight and gains flippable hexagons. It is also the lowest eigenvalue in the fully diagonalized 9104-dimensional 48-site ground-state winding sector, including both spin-reversal parities. We do not elevate that finite-sector statement to a claim about all winding sectors or about the full microscopic spin model. Parity excludes only even photon number; the three-photon channel is excluded separately, and by a wide margin, on kinematic grounds (Sec.), while the flux-correlation diagnostic discussed below disfavors a localized magnetic pair. The combination of collective ancestry, strong finite-momentum softening, and qualitatively different microscopic response motivates the roton-like terminology.

It is tempting to read the census as a sign rule: an excitation disrupts ring resonance, so it should *lose* flippable plaquettes, and a level that gains them is thereby marked as different in kind. That reading is too strong, and the exact statement is worth having because it says precisely how much is true.

The operator $N = \sum_p n_p^{\text{flip}}$ is diagonal in the configuration basis, and each flippable hexagon of a configuration contributes exactly one off-diagonal element of the ring Hamiltonian. Hence $\text{Tr } N$ equals the number of nonzero off-diagonal elements of H , and the census averaged over *all* eigenstates of a sector of dimension D is

$$\frac{1}{D} \sum_n \frac{d\omega_n}{dV} = \frac{\text{nnz}(H)}{D} - \langle 0 | N | 0 \rangle, \quad (\text{S6})$$

which is negative precisely because the ground state is more flippable than a typical ice configuration. On the 48-site cluster Eq. (S6) gives -0.9139 , and direct diagonalization of all 9104 eigenstates returns the same number to 10^{-14} .

So excitations do lose flippability on average, exactly and provably. Nothing, however, constrains an individual level. Of the 9104 eigenstates, 58 — 0.64% — have a positive census, and 20 exceed $+0.20$. The anomalous L level at $+0.339$ sits in that top 0.22% tail, which is the honest form of the statement: it is extreme, not unique.

The census is also the least size-stable of our diagnostics, and we say so plainly. At the anomalous level it

is $+0.339$ on 48 sites, $+0.338$ on 64, $+0.134$ on 72 and $+0.259$ on 96. The sign is robust; the magnitude is not, and no normalization we tried repairs it. On 48 and 64 sites the anomalous level is the only appreciably positive bright census, exceeding the next by factors of 29 and 7; on 72 sites several bright levels are positive and it is no longer the largest. We therefore do not use the census as the primary discriminant.

What does survive on every cluster is a comparison made *within* one of them, and it now appears at two sizes. The 64- and 96-site boxes each carry two inequivalent L -type classes at identical $|\mathbf{q}|$ and identical Gaussian energy. The class that reconstructs has census $+0.338$ (64) and $+0.259$ (96); the class that does not has -0.166 and $-0.02/-0.09$ respectively. Same momentum star, same free-theory prediction, opposite response to the flippability probe. That contrast needs no comparison across cluster sizes and no fitted scale, and it is the census evidence we rely on.

WHY THE RATIO IS SMALL NEAR L : GEOMETRY, AMPLITUDES, AND CONTROLS

Two ingredients coincide at this momentum. The first is geometric. A hexagon contained in a wavefront, i.e. in a plane of constant phase perpendicular to $\mathbf{q}$, encloses zero flux from the electric-field wave: the alternating loop sum defining g_p vanishes exactly. Such a plaquette contributes nothing to the first-moment expression f_1 [Eq. (9) of the main text]. Pyrochlore hexagons are perpendicular to the four $\langle 111 \rangle$ directions, and at the sampled zone-boundary point $L \parallel [111]$ one of the four hexagon families lies in the wavefronts. That family therefore drops out of the oscillator-strength cost exactly, and the vanishing is an identity, not a small number: resolved by family, the geometric factor $\sum_p \sum_\mu |g_p^\mu(\mathbf{q})|^2$ at L is (96, 96, 96, 0), against (96, 96, 96, 96) at X . The reduction of the *geometric* factor is therefore exactly $3/4$. The reduction of f_1 itself is not exactly $3/4$, because the first-moment expression f_1 [Eq. (9) of the main text] weights each plaquette by its own ring coherence $w_p = \langle W_p + W_p^\dagger \rangle$, which is not uniform on an anisotropic cluster; numerically $f_1(L)/f_1(X) = 0.684/0.880 = 0.777$.

It is important that this geometric decoupling is necessary but not sufficient. It occurs at every momentum along $[111]$: at the smallest such momentum the family resolution is (72, 0, 72, 72), again with one family exactly dark. Yet that momentum shows no anomaly, its dominant level sitting at $1.98K$ against a Gaussian value of $1.76K$. What distinguishes L is the second ingredient below.

The second ingredient is dynamical. In the interacting $V = 0$ ground state the equal-time structure factor is largest at L among the sampled momenta: $S(L) = 0.370$, compared with 0.19–0.29 elsewhere. The combination of

reduced f_1 and enhanced S makes f_1/S smallest at L among the twelve cluster momenta and makes it the only sampled point where the single-mode energy falls below the Gaussian-photon line.

Crucially, the enhancement of $S(L)$ is not inherited from the ice constraint alone, and two independent controls establish this.

The first is exact and requires no sampling. The equal-amplitude superposition over the *same* 9104 configurations of the *same* cluster is the classical ice ensemble restricted to exactly what the quantum state can reach, so comparing it with the interacting ground state isolates the effect of the amplitudes alone. It gives $S(L) = 0.297$ against the interacting value 0.370. Measured as an enhancement over the other sampled momenta, the classical ensemble raises L by 11% and the interacting ground state by 48%. Roughly four fifths of the L enhancement is therefore made by the quantum dynamics, not by the ice rule.

The second checks that this is not an artifact of the small cluster. Loop-update Monte Carlo on a 2048-site classical ice manifold, binned into 30 independent blocks, gives $S(L) = 0.2488(67)$, $S(X) = 0.2538(73)$, $S(W) = 0.2475(52)$ and $S(K) = 0.2467(46)$. The total spread across these zone-boundary momenta, 0.007, is comparable to the typical statistical error, 0.006: within the resolution of the calculation the classical structure factor is flat, and L in particular is not distinguished. (The zone center is excluded from this comparison; it is the uniform mode, and it carries no weight in the finite cluster.) The much stronger L enhancement seen in the interacting ground state is generated by the quantum ring dynamics at $V = 0$. This is the main distinction from an RK-point resonon, whose softening can be read directly from the classical correlations of the equal-amplitude state.

Finally, the mechanism is the *ratio*, and L is where the ratio is minimized rather than a privileged point in itself. The 72-site cluster makes this concrete: its momentum $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ has a family decomposition (36, 144, 144, 108) — one hexagon family suppressed to a quarter rather than the exact zero that L enjoys — but a structure factor almost as large, and its ratio comes out within 0.9% of the L value. Its spectrum is correspondingly almost identical: a dominant level at $1.075K$ with 55% of the weight, against $1.068K$ with 54% at L . Exact geometric decoupling is therefore one route to a small ratio; partial suppression combined with a large $S(\mathbf{q})$ is another, and the spectrum follows the ratio, not the route.

FINITE-SIZE AND DEFORMATION CHECKS

Convergence of the collective state. A systematic variational improvement of the electric-field wave converges rapidly to the exact 48-site eigenvalue, $1.450 \rightarrow 1.074 \rightarrow$

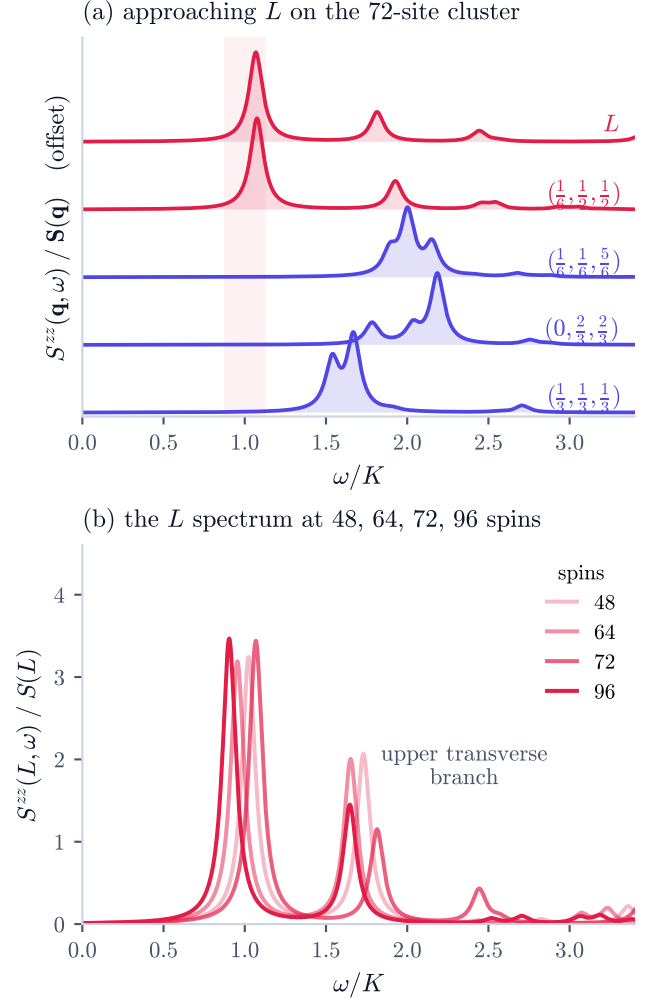


FIG. S1. The mode itself: exact $S^{zz}(\mathbf{q}, \omega)$ near L . (a) The 72-site spectrum at five momenta, each normalized by its own $S(\mathbf{q})$; the poles are Lanczos-converged (sum-rule gate 1.0000), broadened by a display-only $\eta = 0.05K$. At the three photon-like momenta the response peaks at 1.5 – $2.2K$; at L and at the soft-valley momentum $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ (Sec.) it collapses onto a single dominant pole near $1K$ carrying over half the weight (shaded band). (b) The L -point spectrum at all four diagonalized sizes, overlaid: the soft pole sits at 0.90 – $1.07K$ with about half the weight at every size, the upper transverse branch at 1.65 – $1.81K$. Nothing here is fitted or extrapolated; the figure is the spectrum.

$1.034 \rightarrow 1.026K$ against $1.024K$ from complete diagonalization. The state is reached from the probe in a few steps, as a collective mode should be, rather than requiring the full Hilbert space.

A prediction, not a fit. The 64-site cluster provides the one test in which the ground-state ratio is used in the direction it is supposed to work: forward. Its anisotropy splits the L star into two inequivalent classes, which the Gaussian theory cannot distinguish — both have the same $|\mathbf{q}|$ and the same Gaussian energy. They differ in

a ground-state quantity, the equal-time structure factor: $S = 0.369$ for one class and $S = 0.227$ for the other. The f_1/S mechanism therefore predicts, before any excited state is computed, that the first class reconstructs and the second does not. It does. The class with $S = 0.369$ carries a level at $0.954K$ with 50.0% of the weight; the class with $S = 0.227$ has its dominant level at $1.998K$ with 77% of the weight, and no anomalous state at all. The effect follows the structure factor rather than the label L .

Stability with size. We can now compare four clusters. The third, $2 \times 3 \times 3$ with 72 spins, required extending the state representation past a single 64-bit word; the fourth, $2 \times 3 \times 4$ with 96 spins, required in addition a vectorized enumeration of its two-billion-configuration ice manifold and a Lanczos treatment of a $146\,758\,352$ -dimensional winding sector (Appendix). Ninety-six spins matches the largest constrained-space exact diagonalization reported for this model [8], there used for ground-state properties; here the full operator-resolved response is obtained. The four boxes span three distinct shapes.

	$2 \times 2 \times 3$	$2 \times 2 \times 4$	$2 \times 3 \times 3$	$2 \times 3 \times 4$
sites	48	64	72	96
ice configurations	87 546	2.76×10^6	1.28×10^7	2.01×10^9
sector dimension	9 104	249 180	973 008	1.47×10^8
E_0/N	-0.2199	-0.2165	-0.2057	-0.2038
$S(L)$	0.370	0.369	0.321	0.355
ω_{low}/K	1.024	0.954	1.068	0.904
share	50.8%	50.0%	54.0%	54.4%
ω_{high}/K	1.730	1.652	1.814	1.647
$\omega_{\text{low}}/\omega_{\text{high}}$	0.592	0.578	0.589	0.549
census $d\omega_{\text{low}}/dV$	+0.34	+0.34	+0.13	+0.26

The qualitative picture is identical on all four: a single dominant level near $1K$ carrying at least half of the weight at L , and a second transverse branch roughly seventy percent higher. The scale-free ratio of the two branch energies varies over 0.549–0.592 across a factor of 16 000 in sector dimension and three box shapes; the soft level’s share of the weight varies over 50–54%. This is the evidence least contaminated by fitted quantities. On the largest cluster the mode is, if anything, the softest of the four.

The class prediction repeats at 96 sites. The $2 \times 3 \times 4$ box, like the 64-site one, carries two inequivalent L classes, with $S = 0.355$ and $S = 0.241$. Computed before any excited state, the ratio again selects the first: it develops the $0.904K$ level with 54.4% of the weight, while the second remains photon-like, its two dominant levels at $1.651K$ and $1.730K$ with flippability censuses of -0.02 and -0.09 against $+0.26$ for the soft mode. The forward discrimination between L classes has therefore

succeeded at two sizes independently, and the census contrast within a single cluster — positive for the class that reconstructs, negative at the same $|\mathbf{q}|$ for the class that does not — appears at both.

Not everything is stable, and we say which. $S(L)$ moves between 0.321 and 0.370 with box shape; the weight retained by the upper branch varies between 18% and 32%; the census magnitude varies by a factor 2.6 (Sec.); and the polarization splitting at X is not converged, falling from 42% at 48 sites to 5.9% at 64; no conclusion is drawn from it.

The deformation trajectory. Under a bias toward more flippable configurations ($V < 0$), $f_1(L)$ changes by less than one percent while $S(L)$ grows and the level softens from $1.02K$ to $0.73K$. The ground-state factor moves first and the spectrum follows, which is the ordering the f_1/S mechanism requires and the opposite of what a fitted correlation would produce.

THE SOFT VALLEY

A quasiparticle minimum should be concave up, and one should ask in which directions it is, and whether the softness is confined to a single point. The complete grid answers both. Away from L the ratio rises in *every* allowed non-radial direction: to 2.216(5) at $(\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$, to 2.265(5) at $(\frac{3}{8}, \frac{3}{8}, \frac{5}{8})$, to 2.484(5) at $(\frac{1}{4}, \frac{1}{2}, \frac{3}{4})$, and steeply to 2.78 at W — so L is a genuine minimum of the landscape, not a saddle among the zone-boundary momenta. (Along the radial [111] line the point is stationary by band periodicity, as at any zone corner; a lattice minimum lives in the boundary manifold, exactly as the magnon roton-like minima of frustrated antiferromagnets do at their zone corners.)

The rise is, however, strongly anisotropic: the landscape at 2048 sites has a shallow *valley* along the $(t, \frac{1}{2}, \frac{1}{2})$ lines, with the ratio varying by only two percent from $(0, \frac{1}{2}, \frac{1}{2})$ through $(\frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ to its minimum at L , while every direction transverse to the valley climbs by 15–28%. The softness is therefore not an isolated point: it is the bottom of a flat trough. And this is not only a statement about the centroid: the 72-site cluster carries a momentum on exactly this line, $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$, and its *exact* spectrum shows a soft level at $1.075K$ with 55% of the weight — essentially degenerate with the L mode at $1.068K$ (main text). The prediction that the trough hosts soft modes away from L is thus verified spectrally at the one accessible momentum on it. A flat valley of nearly degenerate soft collective modes, rather than a single deep pocket, is itself a statement about the physics: whatever ordering tendency the softening signals is frustrated among the wavevectors of the valley.

FEW-PHOTON KINEMATICS

Spin-reversal parity alone does not identify the low L level. Under charge conjugation an n -photon state carries parity $(-1)^n$, so the odd parity of the level excludes even photon number but permits $n = 3$. We therefore test the three-photon assignment kinematically. Using the Gaussian branch quantized on the same torus (Appendix), we minimize the total energy over every partition of the cluster momentum into n photon momenta drawn from the allowed set, excluding Γ , which carries no photon because the curl has no nonzero singular value there. At L the resulting thresholds are

$$\omega_L^{(1)} = 2.03K, \quad \omega_L^{(2)} = 4.03K, \quad \omega_L^{(3)} = 5.55K. \quad (\text{S7})$$

The level in question lies at $1.02K$, a factor of 5.4 below the three-photon threshold and below even the one-photon energy. The same ordering holds at every sampled momentum: the three-photon threshold never falls below $5.28K$, because the softest photon available on this cluster already costs $1.76K$. A three-photon state built from weakly renormalized Gaussian photons is therefore excluded on the clusters we can diagonalize; since the paper itself establishes that interactions at the zone boundary are order one, this is a statement about that description, not a kinematic theorem about arbitrarily dressed constituents.

This argument uses the Gaussian dispersion to count the cost of the constituents, and so it is an argument about weakly renormalized photons. It does not by itself exclude a multiparticle state built from strongly renormalized constituents. What makes that alternative implausible here is that the level is a single, sharp, dominant pole carrying half of the weight at its momentum rather than the small distributed weight characteristic of a continuum, and that its polarization content is that of a single transverse branch (Appendix).

OFF-POLE WEIGHT AND THE POLARIZATION SPLITTING

The two transverse polarizations are exactly degenerate in the Gaussian theory, and their interaction-induced splitting was predicted in semiclassical treatments [3, 4]. Our calculation determines its magnitude in the microscopic ring model without an expansion. Because the two brightest levels at each momentum are polarization-pure and mutually orthogonal (Appendix), the splitting can be read off directly as the energy difference between them. It grows monotonically as the momentum increases: it is unresolved, below the $10^{-8}K$ level-grouping tolerance, at the smallest momentum of the 64-site cluster, 4.4% at the smallest momentum of the 48-site cluster, 17% at $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$, 42% at X and 51% at L . The degen-

eracy is thus restored in the infrared, as the Gaussian theory requires, and destroyed at the zone boundary.

We emphasize what this does *not* add. The splitting at L and the roton-like softening at L are the same fact: the softened level is the lower of the two transverse branches, and the splitting is the size of the softening. It is reported here because it makes contact with a specific earlier prediction, not as independent evidence for the anomaly.

Beyond the two photon descendants, 16–38% of the S^z spectral weight resides in higher levels absent from the Gaussian one-photon theory. This fraction is defined without any energy window or fitted scale: the two photon descendants are identified as the brightest level in each transverse polarization channel, and the off-pole weight is everything else (Appendix). An alternative operational definition — simply the two brightest levels at that momentum — gives identical numbers at every momentum sampled. The off-pole fraction takes its smallest value, 16% on 48 sites and 13% on 64 sites, at the smallest momentum available on each, consistent with recovery of the infrared photon.

Several diagnostics favor a multiparticle rather than a localized magnetic interpretation of this higher weight. Plaquette-flux correlations are weak and spatially extended instead of showing the localized disturbance expected for a bound magnetic pair, and the response to the RK deformation is consistent with sums of single-mode responses. By contrast, leading-order perturbation theory in the quartic term obtained by expanding the cosine predicts polarization splittings and transferred spectral weight below 0.1% at the coupling generated by this model, and only a few percent even when that coupling is artificially tripled. The finite-momentum spectrum is therefore not quantitatively described as a weak anharmonic correction to the Gaussian photon.

PARITY AND COMPLEMENTARY PROBES

Global spin reversal leaves Eq. (1) of the main text invariant. Eigenstates can therefore be chosen even or odd under this symmetry, while S^z is odd. Consequently a spin-reversal-even ground state couples through neutron scattering only to odd states. This elementary selection rule is exact in the model; numerically every even level carries S^z weight below 10^{-28} . In gauge language it is charge conjugation, under which an n -photon state has parity $(-1)^n$.

The practical implication is a clean division between linear and bilinear probes. Even-sector weight, including the two-photon channel, is invisible to S^z but can appear in probes coupling to spin pairs, such as Raman scattering [9] and ultrasound [10]. Stray-field noise magnetometry [11], which couples to magnetization, belongs to the neutron-like odd channel. The selection rule itself is not a new symmetry statement; its value here is that

the complete finite-size spectrum makes the complementary experimental channels explicit.

EXACT FAMILY DECOMPOSITION OF THE RELAXATION ENERGY

Because the ring term is the entire nonconstant Hamiltonian inside the ice manifold, the excitation energy of any state $|n\rangle$ is exactly a sum of plaquette contributions,

$$\omega_n = -K \sum_p \left[\langle n | W_p + W_p^\dagger | n \rangle - \langle 0 | W_p + W_p^\dagger | 0 \rangle \right], \quad (\text{S8})$$

with no remainder. Grouping the plaquettes by hexagon family — the four $\langle 111 \rangle$ orientations, labeled by the sublattice each hexagon omits — turns Eq. (S8) into a four-term decomposition of any excitation energy.

Two states are compared. The bare single-mode state is the lowest eigenvector of H in the span of the sublattice-resolved probes $S_\mu^z(\mathbf{q}_L)|0\rangle$ with the ground state projected out; that span is exactly two-dimensional at every size, consistent with the rank-two transverse curl of Appendix . (The projection is not optional on fcc(2,3,3): there the L -weighted sublattice-3 sum is a conserved quantity of the ring dynamics — family-3 hexagons contain no sublattice-3 site and the sublattice-3 site pairs of the other families are L -commensurate on that torus — so one probe channel is proportional to the ground state itself, though with negligible norm, so $S(L)$ is uncontaminated.) The exact soft state must be chosen with equal care: the level is degenerate, and the per-family coherences of an arbitrary multiplet member are basis-dependent. We use the basis-invariant choice, the normalized projection of the probe onto the multiplet — the state the probe creates — and verify that the alternative alignment, to the single-mode state, gives per-family values identical to four decimals. Three gates hold at every size: the probes are orthogonal to the ground state at the 10^{-15} level, the multiplet captures exactly the level's known share of $S(\mathbf{q}_L)$, and the projected state's energy reproduces the soft eigenvalue to six decimals.

The result, in units of K , with the geometrically dark family (the one with $g_p(\mathbf{q}_L) \equiv 0$) marked $\star$:

	48	64	72
single-mode energy	1.450	1.412	1.481
exact soft level	1.024	0.954	1.068
recovery	29%	32%	28%
relief, family 1	+0.080	+0.040	+0.276
relief, family 2	−0.064	−0.036	+0.002
relief, family 3	+0.088 $\star$	+0.227	+0.002
relief, family 4	+0.322	+0.227	+0.133 $\star$
dark-family share	21%	9% $\star$	32%

(families are listed in a fixed per-cluster order; at 64 the dark family is family 1). The totals reproduce the variational and exact energies exactly, so the decomposition is complete rather than approximate.

Two conclusions follow. First, the 28–32% variational recovery is stable across all three sizes: the collective mode robustly lowers its energy below the single-mode bound by a correlated, family-selective reorganization of ring coherence. Second, the relief is *not* concentrated in the geometrically dark family: its share is 9–32% with no size trend, and the largest single-family share resides in bright families whose pattern varies with cluster geometry. An earlier version of this analysis reported 71% from the dark family; that number was an artifact of decomposing one arbitrarily chosen member of the degenerate multiplet, and the Feynman–Cohen backflow interpretation built on it is withdrawn. We record the correction explicitly because the basis-dependence of degenerate-multiplet expectation values is an easy trap in this kind of analysis.

GREEN'S-FUNCTION MONTE CARLO

The ring Hamiltonian restricted to the ice manifold is stoquastic: its diagonal vanishes and every nonzero off-diagonal element equals $-K$. The power method with the propagator $\Lambda - H$, Λ larger than the top of the spectrum, therefore converges to the ground state with strictly nonnegative amplitudes, and importance sampling with the equal-amplitude trial state $|\Psi_T\rangle = \sum_x |x\rangle$ gives walkers whose weights never change sign. A walker at configuration x moves to one of its $n^{\text{flip}}(x)$ ring-connected neighbors or stays, with the local weight $\Lambda - (\text{diagonal}) = \Lambda$ and off-diagonal flow $K n^{\text{flip}}(x)$; population control uses comb reconfiguration at fixed walker number, and the mixed estimator gives $E_0 = -K \langle n^{\text{flip}} \rangle_{\text{mix}}$.

$S(\mathbf{q})$ is diagonal in the configuration basis, and for diagonal observables the mixed distribution $\Psi_0 \Psi_T$ is converted to the pure one $|\Psi_0|^2$ by forward walking: an observable measured at time t is weighted by the number of descendants its walker retains at $t + T$. Convergence in T and in the walker number is quantified against the exact 48-site reference by a scan over 256–2048 walkers and $T = 20$ –160: at $T = 20$ the forward walking is demonstrably under-converged, biasing $S(L)$ low by about 0.7% (five standard deviations at our best statistics), while for $T \geq 40$ every entry scatters about the exact 0.370365 within errors; production runs use $T = 80$. The mixed-estimator energies in the same scan sit $\approx 1.5 \times 10^{-4}$ (relative) above the exact ground energy independently of walker number, a residual population-control bias two orders of magnitude below any energy difference used in this paper.

Initialization has one subtlety worth recording. Every

sublattice-uniform ice state is a frozen point of the dynamics — each hexagon meets its three sublattices twice with opposite alternation parity, so no hexagon is flip-pable — and it is also maximally polarized, so its winding sector contains no other state and sector-preserving loop updates cannot leave it. The seed is therefore randomized by worm-constructed loop flips in two phases: first accepting loops that monotonically reduce the integer polarization $\sum_i (2n_i - 1) 4\mathbf{r}_i$ (with a brief relaxation of monotonicity when stalled), then, once the minimal reachable polarization is attained, loops that preserve it exactly. Not every box reaches zero: on fcc(2, 3, 3) the zero-polarization sector is empty, a fact known exactly from the 72-site diagonalization, and the energy gate below is what certifies the sector reached.

Every production number is gated against exact diagonalization. On the 48-site cluster the method reproduces $E_0 = -10.556012$ to 0.5σ and $S(L) = 0.370365$ to every printed digit; the equal-amplitude value of $S(L)$ is 0.297, so a defective forward walker could not pass. On the 72-site cluster it reproduces $E_0/N = -0.205690$ to 1.8σ in the correct nonzero-polarization ground sector. Statistical errors are from binned averages (16 bins); f_1 carries no statistical error beyond that of E_0 , since on cubic supercells the sum rule fixes the uniform plaquette coherence as $w_p = -E_0/(K n_{\text{hex}})$ and the remaining fac-

tor is exact geometry. Production runs used 1024 walkers and 30 000–60 000 steps; the 2048-site full-grid run measures $S(\mathbf{q})$ at all 512 allowed momenta as a single matrix product per snapshot.

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- [1] T. Fennell, P. P. Deen, A. R. Wildes, K. Schmalzl, D. Prabhakaran, A. T. Boothroyd, R. J. Aldus, D. F. McMorrow, and S. T. Bramwell, *Science* **326**, 415 (2009).
 - [2] O. Benton, O. Sikora, and N. Shannon, *Phys. Rev. B* **86**, 075154 (2012).
 - [3] M. P. Kwasigroch, B. Douçot, and C. Castelnovo, *Phys. Rev. B* **95**, 134439 (2017).
 - [4] M. P. Kwasigroch, *Phys. Rev. B* **102**, 125113 (2020).
 - [5] M. Hermele, M. P. A. Fisher, and L. Balents, *Phys. Rev. B* **69**, 064404 (2004).
 - [6] D. S. Rokhsar and S. A. Kivelson, *Phys. Rev. Lett.* **61**, 2376 (1988).
 - [7] N. Shannon, O. Sikora, F. Pollmann, K. Penc, and P. Fulde, *Phys. Rev. Lett.* **108**, 067204 (2012).
 - [8] S. D. Pace, S. C. Morampudi, R. Moessner, and C. R. Laumann, *Phys. Rev. Lett.* **127**, 117205 (2021).
 - [9] Z. Fu, J. G. Rau, M. J. P. Gingras, and N. B. Perkins, *Phys. Rev. B* **96**, 035136 (2017).
 - [10] E. Z. Simon, A. S. Patri, and Y. B. Kim, *Phys. Rev. B* **106**, 064427 (2022).
 - [11] G. K. Naik, J. N. Hallén, N. C. Jayarama, R. Moessner, and C. R. Laumann, *Phys. Rev. Lett.* (2026), arXiv:2512.14843.

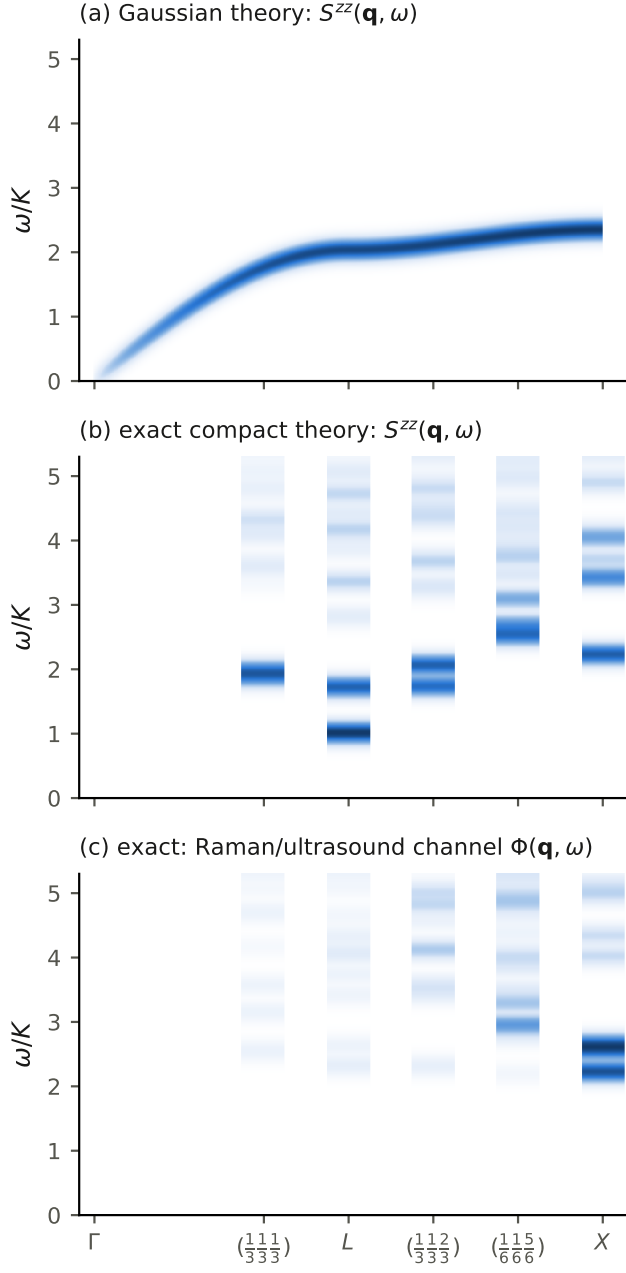


FIG. S2. Simulated spectra along Γ - L - X for the 48-site cluster, broadened by $\eta = 0.08K$ and normalized separately in each panel. (a) Gaussian one-photon neutron response. (b) Exact operator-resolved S^z response within the ground-state winding sector: split photon descendants, a dominant low L mode at about half the Gaussian-photon energy, and higher off-pole weight. (c) Spin-reversal-even response accessible to bilinear probes. By the parity selection rule no eigenstate is shared between panels (b) and (c).